CHRISTIAN ALEMÁN PERICÓN

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WORK EXPERIENCE

Micro1 Inc. — **San Francisco, USA (Remote)**
Economics Research, AI Training — **2026 -Present**

- Provide expert insights on automation using Eviews, Stata, Python, R, Matlab.
- Provide professional advisory services on macroeconomic theory and interpretation, finance, econometrics, price theory, economic research design, economic model diagnostics and validation, forecast evaluation, model benchmarking and peer-review publication.
- Conduct AI training. Advise on the institutional design for AI-enhanced platforms.

New York University Abu Dhabi — **Abu Dhabi, UAE**
Research Associate, Economics Division — **2023 – 2026**

- Developed heterogeneous agent macroeconomic models to inform fiscal and monetary policy using MATLAB and Python.
- Led data intensive projects integrating causal inference and forecasting techniques for real time decision making.
- Designed and implemented a rigorous evaluation framework to assess the ex-ante/ex-post economic impact of the 2024 French payroll subsidy reform, using a structural DGE macro-model calibrated on confidential administrative microdata.
- Designed optimal tax schemes using government data to regulate the purchase of high-cost health treatments in Denmark.
- Applied machine learning methods (Least Squares Learning, Self Fulfilling Equilibria) and deep learning architectures to large-scale longitudinal data sets, developing benchmarks to assess forecasts of aggregate labor market expectations and health outcomes.
- Applied deep learning architectures (CNNs, GANs, Autoencoders) to solve structural decision models, analyzing model behavior and failure modes to guide research priorities.
- Delivered 15+ academic talks and policy presentations across 10+ countries showcasing the above listed projects.
- Authored research papers on causal inference, quantitative policy evaluation, and AI-driven decision systems for top-tier peer-reviewed journals (AEJ Macroeconomics, Econometrica, Review of Economic Studies), contributing to the development of novel evaluation methodologies.

Université Libre de Bruxelles (ULB) – ECARES — **Brussels, Belgium**
Research Fellow, Economics Department — **2022**

- Applied econometric methods (Generalized Method of Moments, Monte Carlo simulations, Non-parametric Estimation) using MATLAB to study income and wealth inequality resulting from intergenerational transfers, bequests and migration choices.
- Developed a DSGE macro-model to study the effects of housing market shocks on household consumption and savings.

Barcelona School of Economics — **Barcelona, Spain**
Predoctoral Researcher — **2018 – 2022**

- Co-developed and published a novel econometric evaluation methodology (Stage-Based Identification) to estimate real-time effects of nationwide policies across health, environment, and labor domains; routines publicly available in R, MATLAB, and Python.
- Constructed a multi-phased dynamic macro-model in MATLAB to estimate aggregate effects of HIV in developing economies.
- Developed a framework to identify match quality in labor markets using revealed preferences, improving match predictions by 40%.
- Led an Advanced Research Methods course for 40 students, focusing on data analysis using Python and R.
- Managed projects for governments, think tanks and corporations, aligning research outputs with stakeholder's strategic priorities.

Bank Sabadell — **Barcelona, Spain**
Research Assistant, Department of Quantitative Analysis — **2018**

- Developed and benchmarked time series models (VAR, ARCH, GARCH) to forecast key macroeconomic and financial variables (inflation, GDP, credit) under risk scenarios, producing structured performance assessments for internal strategy teams.
- Estimated a Vector Error Correction model to monitor and forecast credit card consumption trends in Mexico.
- Evaluated macro-prudential and monetary policy using agent based models and leading indicators for industrial output and GDP.
- Collaborated with risk management, strategy and business intelligence teams to align economic outlooks with business goals.

Halle Institute for Economic Research — **Halle (Saale), Germany**
Research Intern, Department of Macroeconomics — **2017**

- Developed a DSGE input-output model to identify the determinants of long-term growth processes in Germany.

EDUCATION

Universitat Autònoma de Barcelona, University of Pennsylvania — **Spain, USA**
Ph.D. Economics (Cum Laude) — **2018 – 2022**

Université Paris I Panthéon-Sorbonne — **France**
M.Sc. Master in Models and Methods of Quantitative Economics — **2016 – 2018**

AWARDED FUNDING

Ludwig Maximilian University of Munich and NYU Collaboration Grant — **2025**
New York University Postdoctoral Research Fund Award — **2023 – 2026**

SKILLS / ADDITIONAL INFORMATION

Programming & Tools: Python, Matlab, R, Julia, Stata, Eviews, SQL, HPC, AWS, Bloomberg, ML/AI: PyTorch/TensorFlow familiarity
Linguistic Proficiency: Spanish (Native), English/French/German (Fluent), Italian/Catalan (Intermediate)
Citizenship: Spain, Canadian Permanent Resident

SELECTED RESEARCH

The AI Economy: Adoption, Human Validation, and Labor Demand (2026) ***Submitted: Journal of Monetary Economics***

I study how human validation requirements shape the response of wages and labor demand to AI adoption. I introduce fixed, recurring validation labor into a general equilibrium task model with endogenous AI adoption. It features a validation threshold determining whether the labor created by AI adoption offsets the productive labor it displaces after accounting for demand expansions. I estimate the model using U.S. occupation-task data on human versus AI involvement at work, and AI adoption rates. The estimated U.S. validation requirement lies below the threshold, implying that productive labor displacement dominates and wages fall. For the occupations in the estimation sample, a ten year projection of declining AI costs raises adoption from 18 to 62 percent and nearly triples output, while reducing employment by 13 percent and wages by 43 percent. Extending the exercise to the entire U.S. economy attenuates the employment and wage declines to 5 and 18 percent, respectively.

Kramer vs. Kramer. On the importance of Children and Divorce Filings for Understanding Divorce Rates in the USA (2026) ***American Economic Journal: Macroeconomics, R&R***

I document that approximately 70% of divorce filings in the 1970's were done by wives in the United States. Since then, this figure has experienced a large decline, reaching 56% in 2015. At the same time, divorce rates sharply increased from 1960 until the mid 1980's and have declined since then. I construct a life cycle model of endogenous marriage and unilateral divorce with endogenous labor supply and savings that jointly explains these facts. I use my model to measure the contribution of changes in the gender-wage gap, property division laws and child custody arrangements in explaining the divorce patterns over time. First, the reduction in the gender-wage gap generates two opposing effects. On the one hand, the reduction of the gender-wage gap increases the value of divorce for married women and, on the other hand, unmarried women become more selective in the marriage market thus raising the quality of newly formed matches. Second, children increase the value of divorce for the custodial parent; so a higher probability of getting child custody raises the chances of filing for divorce. Third, a higher share of assets assigned to wives upon divorce can either increase or decrease divorce rates by altering the savings decision of the household. My model accounts for approximately 50 per cent of the decline in divorce filings and 70 per cent of the variation in divorce rates between 1970 and 2015. I find that the decrease in the gender-wage gap and the increase in the probability of getting child custody for men are major drivers behind the changes in divorce rates and in divorce filings, respectively. Importantly, I find that failure to match who files for divorce can lead to opposite counterfactual results.

The Ozempic Economy: Sugar Addiction, Labor Market and Optimal Subsidies (2025) with Fane Groes, Veda Narasimhan, Raul Santaeulàlia-Llopis and Tetyana Surovtseva.

Using linked Danish administrative data on medical prescriptions and firm records, we examine the effects of weight-loss pharmacotherapy — such as Ozempic and Mounjaro — on employment, hours, and labor earnings. Our contribution is twofold. First, we document empirically the significant and robust effects of weight loss drugs on labor market outcomes using a battery of alternative identification strategies in diabetic and non-diabetic population. Second, we build a quantitative macroeconomic model with endogenous sugar consumption and pharmacotherapy uptake to assess optimal public subsidies and alternative regulatory frameworks such governing pharmaceutical pricing.

Employment and Human Capital Investment Responses to Payroll Tax Reductions (2025) joint with Etienne Wasmer, Pranav Mimani

To counter the negative effects on employment for low-skilled workers, some countries have developed policies that reduce social contributions for low wage workers. We examine the effects of such policies on employment, the incentives to invest in employee training, and salary progression. We propose a model featuring dual heterogeneity of workers and firms, as outlined by Shimer and Smith (2000), to which we add specific human capital accumulation. We calibrate the model on France, where the policy of tax relief, initiated in 1993, has reached a record of 80 billion euros (2.5% of GDP). We find that the reductions in social contributions for low wage workers disincentivize human capital investments and contribute significantly to the high concentration of workers who prefer to keep earning the minimum wage. Our simulations suggest that a one percent reduction of the social contribution subsidy increases average wages by 4% and net employment by 2.5%.

A Quantitative Theory of the HIV Epidemic: Education, Risky Sex and Asymmetric Learning (2024) joint work with Daniela Iorio and Raül Santaeulàlia-Llopis ***The Review of Economic Studies, R***

We explore learning about HIV infection odds from risky sex as a new mechanism explaining the Sub-Saharan Africa HIV epidemic. Our novel empirical evidence reveals a U-shaped relationship between education and being HIV positive across the epidemic, which prompts the idea of asymmetric learning: more educated individuals potentially learn faster and update their (latent) beliefs about infection odds more accurately than less educated individuals, inducing earlier sexual behavioral change among the more educated. Our non-stationary model incorporates three HIV epidemic stages, chronologically: a myopic stage where agents are unaware of how risky sex causes infections, a learning stage where agents update beliefs on infection odds, and an ART stage reflecting treatment introduction. Anchored to the micro evidence— the U-shaped HIV-education gradient—we find that our learning mechanism is powerful: a 5-year earlier learning reduces new AIDS deaths by almost 45%, and a 10-year earlier learning results in a 60% drop. Removing learning altogether reveals that under full information, i.e life expectancy asymmetries present only through education, play a much smaller role than learning.

Stage-Based Identification of Policy Effects (2023) joint with Christopher Busch, Alexander Ludwig and Raül Santaeulàlia-Llopis ***Econometrica, R***

We develop a method that identifies the effects of nationwide policy, i.e., policy implemented across all regions at the same time. The core idea is to track outcome paths in terms of stages rather than time, where a stage of a regional outcome at time t is its location on the support of a reference path. The method proceeds in two steps. First, a normalization maps the time paths of regional outcomes onto the reference path—using only pre-policy data. This uncovers cross-regional heterogeneity of the stage at which policy is implemented. Second, this stage variation identifies policy effects inside a window of stages where a stage-leading region provides the no-policy counterfactual path for non-leading regions that are subject to policy inside that window. We assess our method's performance with Monte-Carlo experiments, illustrate it with empirical applications, and show that it captures heterogeneous policy effects across stages.